State Methodology (10/22)

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Changed on 2 November: 3.4 (baseline preferences fit by target-to-model ratios), 4.2 (Mississippi home capacity), 4.3 (local non-labor residual), 6 (price validation; open items, with the five items from the 2 November review), and notation and wording throughout.

This appendix explains how I adapt the Budget Lab's national ECEC model to local markets. I use the national methodology's definitions and notation. The baseline is 2019. Each run covers children ages 0–4 in one of the 50 states, DC, or New York City (NYC). NYC is the five boroughs; it is also included in New York State, so I exclude its separate run when adding up national totals.

I keep the nationally estimated forests, utility function, β, ρ, production technology, and care-worker labor-supply elasticities. I change local care-hour targets, wages, staffing requirements, non-labor costs, and parents' employment paths. Each geography is a separate market, with no families or workers moving across its boundary.

1 Income ranks and national calibration

1. **Put incomes on a comparable purchasing-power basis.** I divide 2019 household income by the 2019 BEA all-items regional price parity (RPP), expressed relative to the US average of 100:

$$\text{income used for ranking} = \frac{\text{dollar income}}{\text{all-items RPP}/100}.$$

   An income of \$55,000 in a place with RPP 110 becomes \$50,000 for ranking, because the same dollar income buys less in a more expensive place. The rank enters only the prediction models; the household budget uses dollar income.

2. **Assign the available geographic price index.** ACS households use their state's 2019 BEA all-items RPP. NSECE identifies only four Census regions, so each NSECE household uses its region's average state RPP, weighted by 2019 ACS children under five. NYC uses the New York–Newark–Jersey City metropolitan RPP, 114.9, as a proxy for city prices; the metro area includes parts of New Jersey and Pennsylvania.

3. **Calculate national income ranks within each dataset.** I calculate the 1st through 99th percentile cutoffs of adjusted household income among children under five with positive household income, using child weights. I pool all regions for these cutoffs and calculate them separately in NSECE and ACS. A rank therefore compares a family with families nationally, not only in its own state. The 2019 ranks stay fixed for the trained forests.

4. **Calibrate the utility parameters nationally.** After this income adjustment, I estimate one β and one ρ, shared across geographies and one-child and two-or-more-child parent units. The 19 October 2026 calibration gives $\beta = 0.0003405$ and $\rho = 0.0262383$.

Aggregate elasticity	Target	Achieved
Employment with respect to paid care cost	−0.15	−0.1500
Work hours with respect to after-tax income	−0.05	−0.0500
Paid care hours with respect to paid care cost	—	−1.5137
Employment with respect to caregiver wages	—	1.5636

I calculate these responses using parent-unit weights; the paid-care-hours response also weights each unit by its expected care cost. The care-cost experiment raises paid care costs by 1%; the income experiment adds 1% of each family's expected after-tax income as a lump-sum transfer. The first two are calibration targets; the other two are implied by the model.

2 Center price tiers

1. **Define 36 cutoffs.** I use four Census regions, three community density groups, and three child age bands: 0–1, 2–3, and 4–5. States in the same region use the same cutoff for a given density and age. The same hourly price means a different position in the local market across regions, densities, and ages, so the cutoffs vary. Price is a proxy for quality.

2. **Measure each center's price and enrollment.** For 2019 NSECE center-based providers, I divide the standard weekly posted rate at each single year of age by 40. Within each two-year age band, I take the simple mean of the available hourly rates. Only positive prices enter the medians. The median weight is the provider's survey weight times full-time-equivalent enrollment in that age band. Full-time enrollment counts as one and part-time enrollment as one-half; I use the mean of the available single-age full-time shares within the band.

3. **Combine regional and density differences.** Let $\tilde{p}_{r,d,a}$ be the weighted median posted price among centers in region r, density group d, and age band a, with $r = \text{US}$ for all regions pooled. The cutoff is the regional high-density median times the national ratio of density d to high density:

$$\tau_{r,d,a} = \tilde{p}_{r,\text{high},a} \, \frac{\tilde{p}_{\text{US},d,a}}{\tilde{p}_{\text{US},\text{high},a}}.$$

   I use the national density ratio because the regional rural cells contain only 5–35 surveyed centers per age band, and moderate-density cells contain 28–134 (unweighted counts of centers).

4. **Assign each care arrangement to a tier.** A center uses its own region and density group. A family uses its household's region and density group and its child's age band. I compare posted hourly prices for providers and weekly out-of-pocket payment divided by weekly care hours for households. A price at or above $\tau_{r,d,a}$ is high-price. Household payments are net of third-party support, so their tier can differ from the provider's posted-price tier.

NSECE community density measures the urban share of the nearby population, using tracts intersecting a two-mile circle around the anchor tract's population centroid and inverse-distance weights. High density means at least 85% urban, moderate means 30% to under 85%, and rural means under 30%.

Region	Child age	High density	Moderate	Rural
Northeast	0–1	6.73	6.40	5.09
	2–3	5.75	5.30	3.85
	4–5	5.33	4.52	3.58
Midwest	0–1	5.88	5.60	4.44
	2–3	4.75	4.38	3.18
	4–5	4.38	3.71	2.94
South	0–1	4.49	4.27	3.39
	2–3	3.94	3.63	2.64
	4–5	3.73	3.16	2.51
West	0–1	6.36	6.05	4.81
	2–3	5.29	4.88	3.55
	4–5	4.89	4.15	3.29

Cutoffs are in 2019 dollars per hour, rounded to cents.

3 Local care-hour targets

3.1 Children in regular care

1. **Estimate the state participation rate.** I pool the 2016–2019 NSCH observations for children ages 0–4 with a valid yes/no answer on at least ten weekly hours of non-parental care. The rate is the sum of child survey weights for yes answers divided by the sum for all valid answers. Pooling four pre-pandemic years reduces sampling noise in state estimates. NYC uses New York's rate because NSCH does not identify the city.

2. **Convert the rate to children using the model's care definition.** I multiply the rate π_g by the Census 2019 population ages 0–4, Pop_g (county estimates summed), and by a national factor that converts the NSCH ten-hour definition to NSECE's regular-care definition, which starts at five hours. The factor is the 2019 NSECE child-weighted count of children ages 0–4 in regular non-parental care, excluding school programs, divided by the count with at least ten weekly hours of that care:

$$K_g = \text{Pop}_g \, \pi_g \, (1.098054),$$

where K_g is the number of children in regular care. For the 50 states and DC, the result is 11.6 million children in regular care, 59.2% of 19.6 million children ages 0–4. These are population estimates, not unweighted survey counts.

3.2 Children in unpaid centers

1. **Pre-K.** I take the larger of public-school pre-K enrollment from the Common Core of Data and all-settings state pre-K enrollment from Table 2 of the 2019 NIEER yearbook. NYC uses its own all-settings city count. The larger count covers public-school programs outside the state program and state programs outside public schools, without double counting their overlap.

2. **Head Start and Early Head Start.** I convert funded slots to point-in-time enrolled children using national 2018–19 Office of Head Start enrollment-to-slot ratios: about 0.957

for Head Start and 0.923 for Early Head Start. For Early Head Start, I first remove the 3.592% share serving pregnant women. I remove home-visiting and family-child-care options using national program shares of about 1.94% for Head Start and 38.0% for Early Head Start.

3. **Remove children outside the modeled ages.** National 2019 NHES child-weighted shares, with age measured on December 31, remove 24.40% of pre-K children and 10.74% of Head Start children as age five or older. Early Head Start needs no age-five adjustment. NYC's pre-K calendar-year eligibility already keeps its count within ages 0–4, and NYC Head Start serves no age-five children, so both NYC shares are zero. For New York State, I apply the national pre-K and Head Start shares only to the enrollment outside NYC.

4. **Remove pre-K/Head Start overlap.** I calculate one pooled overlap share o: the Head-Start-slot-weighted mean of the overlap shares in the 10 state/DC geographies that use the all-settings pre-K count and have NIEER unduplicated counts (AL, CT, DC, GA, IA, MA, MD, NC, NJ, and NY). I apply this pooled share, $o = 27.52\%$, to adjusted Head Start in every geography:

$$K_{g,\mathrm{UC}} = \text{adjusted pre-K}_g + (1 - o)\text{adjusted Head Start}_g + \text{adjusted Early Head Start}_g,$$

where $K_{g,\mathrm{UC}}$ is the number of children in unpaid centers. State counts alone cannot identify a child in both programs, so the pooled share estimates the overlap. $K_{g,\mathrm{UC}}$ can still overstate the model's unpaid-center category when programs charge fees or children also attend paid care.

3.3 Allocate children and convert to annual hours

1. **Predict the local care mix.** I apply the national model to the geography's ACS families. For each care sector s, I sum each child's probability of that sector times the child weight to obtain $B_{g,s}$. I exclude parental care and combine the two paid-center tiers into one sector for this step. These are predicted population counts of children.

2. **Allocate the remaining children.** After setting unpaid-center children to $K_{g,\mathrm{UC}}$, I distribute $K_g - K_{g,\mathrm{UC}}$ across paid center, paid home, unpaid home, other paid, and other unpaid care in proportion to their B values:

$$K_{g,s} = (K_g - K_{g,\mathrm{UC}})\frac{B_{g,s}}{\sum_{s'} B_{g,s'}},$$

where s' runs over these five sectors. The participation rate and program counts set the level of care; the national model's predictions for local families set the split across sectors.

3. **Calculate six annual-hour targets.** For each sector, including unpaid centers, I use the predicted part-time share $q_{g,s} = B_{g,s,\mathrm{PT}}/(B_{g,s,\mathrm{PT}} + B_{g,s,\mathrm{FT}})$, where $B_{g,s,\mathrm{PT}}$ and $B_{g,s,\mathrm{FT}}$ split $B_{g,s}$ by care hours. At 10 weekly hours for part-time care or 40 for full-time care, for 50 weeks, the target in annual child-hours is

$$\bar{Q}_g^s = K_{g,s}\,[500 q_{g,s} + 2{,}000(1 - q_{g,s})].$$

The model's hour grid makes the targets comparable with simulated care hours.

3.4 Fit local baseline preferences

1. **Express the national preferences at local 2019 prices.** The national model backs out each family's intercepts from its baseline probabilities P^0_{ij}, with $\alpha_{i1} = 0$: $\alpha_{ij} = (\ln P^0_{ij} - \ln P^0_{i1}) - (\tilde{V}_{ij} - \tilde{V}_{i1})$, where $\tilde{V}_{ij} = V_{ij} - \alpha_{ij}$ is the financial part of utility. I apply the same formula with care costs at the geography's 2019 supply prices, $\tilde{V}^g_{ij}$:

$$\alpha^g_{ij} = (\ln P^0_{ij} - \ln P^0_{i1}) - (\tilde{V}^g_{ij} - \tilde{V}^g_{i1}).$$

   At local prices, these intercepts give back P^0_{ij}.

2. **Scale choices by target-to-model ratios.** Each care sector gets a ratio: its local hours target $\bar{Q}^s_g$ divided by the model's hours. Each caregiver group (sex by under 25, 25–34, 35–44, and 45+) gets an employment ratio: the target employment odds divided by the model's. An option's ratio r_{ij} is the product of its children's sector ratios, times the employment ratio if the caregiver works. I multiply each probability by its option's ratio and renormalize:

$$P^g_{ij} = \frac{P^0_{ij}\, r_{ij}}{\sum_m P^0_{im}\, r_{im}}.$$

   Parental care has ratio one, and the two paid-center tiers share one ratio. Renormalizing moves the other shares, so I multiply each ratio by its new target-to-model ratio and repeat until care hours match the six targets and employment matches each group's mean employment-forest prediction. Every family in the geography gets the same ratios, because local data give totals only, not how choices differ by income, age, or family composition. Those differences come from the national model.

3. **Store the fit as intercepts.** Multiplying a probability by r equals adding $\ln r$ to its intercept, so

$$\alpha'_{ij} = \alpha^g_{ij} + \ln \frac{r_{ij}}{r_{i1}}.$$

   Dividing by option 1's ratio keeps $\alpha'_{i1} = 0$; option 1 is full-time high-price center care with the caregiver not employed. The log of the employment ratio is a utility shift on working options, the same form as the national Δ_t. β and ρ stay national.

4. **Use the fitted baseline for policy comparisons.** I hold these preferences fixed when policy changes. Because I fit hours, child counts and part-time shares need not equal those used to build the targets. The model determines the split between low- and high-price centers.

4 Local supply inputs

4.1 Care-worker wages

1. **Fix the occupation mix by worker education.** I use national 2015–2019 ACS workers living in households, with positive earnings and usual work hours. Childcare workers exclude private-household employment; teachers and administrators must work in child day-care services. Preschool special-education teachers count as preschool teachers. Person-weighted shares of childcare workers, preschool teachers, and administrators are 72.6%, 24.2%, and 3.2% without a BA, and 42.6%, 45.8%, and 11.6% with a BA.

2. **Calculate a local wage index.** I take the weighted mean of BLS OEWS median hourly wages for childcare workers (39-9011), preschool teachers (25-2011), and preschool/daycare

administrators (11-9031), using the fixed ACS weights. $R_{g,e}$ is this local mean divided by the national mean, for geography g and education group e. The local 2019 wage is the national 2019 model wage times $R_{g,e}$:

$$w_{g,e,2019} = w_{\mathrm{US},e,2019} R_{g,e}, \qquad w_{\mathrm{US},e,2019} = \begin{cases} 12.06 & \text{without BA,} \\ 19.70 & \text{with BA.} \end{cases}$$

Fixed occupation weights keep differences in occupation mix out of the index. NYC uses the New York metro area, including its New Jersey and Pennsylvania parts.

3. **Project wages.** I multiply the local 2019 wage by national nominal hourly-wage growth and by the change since 2019 in $A_{g,e,y}$, the annual relative wage. A uses 2019–2025 BLS data and the same weights as R, but drops an occupation from the whole path, locally and nationally, if BLS suppresses it in any year, and renormalizes the other weights. $A_{g,e,2019}$ can therefore differ from $R_{g,e}$; only its change enters the wage. After 2025 I hold the relative wage fixed and continue national wage growth. BLS's 2021 methodology change limits comparisons across years.

4.2 Staff hours per child-hour

1. **Select the rule in force on 1 October 2019.** For centers, I record each geography's full table of children per adult, counting required teachers and aides, by age band from birth to 60 months. I use the ordinary band: the ratio that applies to a child of that age by default. I exclude opt-in groupings, such as special mixed-age groups, toddler extensions that need parent agreement, Montessori or half-day ratios, and two-adult group bonuses. Where bands overlap, a month takes the band that starts latest at or before it. For homes, I use one rule in every geography: the largest number of children ages 0–4 that one adult may care for when an infant is present, excluding school-age-only additions. I never use a configuration that needs a second adult or that excludes infants. NYC centers use the city Health Code, NYC homes use the New York State rule, and the other geographies use state rules. These legal limits measure differences in required staffing, not providers' observed staffing choices.

2. **Weight each year of age by month.** Children per adult at age a, $c_{g,a}$, is the harmonic mean of the band ratios over months $12a$ to $12a + 11$:

$$c_{g,a} = \left(\frac{1}{12} \sum_{m=12a}^{12a+11} \frac{1}{\mathrm{ratio}_{g,m}} \right)^{-1}.$$

This handles band cutoffs inside a year. Texas allows 5 children per adult at 12–17 months and 9 at 18–23 months, so its 1-year-olds get 6.43.

3. **Combine New York State and NYC rules.** The New York run includes NYC. I average New York State and NYC staff per child ($1/c$) at each age, using NYC's share of New York children ages 0–4, 46.47% (Census 2019 county population estimates). The NYC run keeps the city's own rules.

4. **Average the center requirements over ages.** Required staff per child averages $1/c_{g,a}$ over ages, weighted by national 2019 NSECE center enrollment ν_a: 426,000, 718,000, 1,010,000, 2,170,000, and 2,340,000 children at ages 0 through 4 (weighted counts):

$$\sigma_{g,\mathrm{center}} = \frac{\sum_{a=0}^{4} \nu_a / c_{g,a}}{\sum_{a=0}^{4} \nu_a}, \qquad \sigma_{g,\mathrm{home}} = 1/\text{selected home capacity}_g.$$

Common age weights make the index reflect rules rather than differences in the ages of children served across states.

5. **Scale the national labor requirements.** The center index divides $\sigma_{g,\text{center}}$ by its average across the 50 states and DC, $\bar{\sigma}_{\text{center}}$, weighted by Census 2019 population ages 0–4 (NYC excluded):

$$I_{g,\text{center}} = \sigma_{g,\text{center}}/\bar{\sigma}_{\text{center}}.$$

I multiply national staff hours per child-hour by the index, for both education groups. The center index applies to low-price, high-price, and unpaid centers; the home index applies to paid homes. An index of 1.2 raises the labor requirement by 20%. Center indices run from 0.715 (Georgia) to 1.351 (North Dakota).

6. **Blend the home index by the licensed share.** Licensing rules apply only to licensed homes. Licensed homes hold $\lambda = 25.7\%$ of children ages 0–5 in paid home-based care (2019 NSECE home-based provider survey, weighted). Unlicensed paid home care keeps the national staffing level:

$$I_{g,\text{home}} = \lambda \frac{\sigma_{g,\text{home}}}{\bar{\sigma}_{\text{home}}} + (1 - \lambda).$$

For comparison, the share is 24.3% for paid children of any age and 20.9% for all children. Home indices run from 0.873 (South Dakota) to 1.135 (Arizona and Rhode Island).

7. **Keep the scope of home capacity explicit.** Selected home capacity is a rough staffing measure: its inverse equals staff per child only when one adult works at full capacity. Arizona uses four certified paid places and Louisiana uses six certified places. Nevada's rule does not set the capacity of a mixed group with an infant, so I keep its 6. Mississippi sets a mixed group's ratio by its youngest child: 4 with a child under 1, 8 with a youngest child age 1. I use 8 (home index 0.940). With 4, its home labor cost is above its NDCP home price, which gives a negative non-labor residual, and its rule is stricter than every other state's. Age limits, own-child counting, and assistant requirements are recorded with each selected input. I do not separately weight municipal rules within a state, apart from New York.

4.3 Non-labor costs

1. **Start with each national residual.** In the national model, the per-unit residual revenue $\bar{\delta}$ of a sector is its 2019 NSECE price minus its labor cost: \$0.510055 per child-hour for low-price centers, \$4.614567 for high-price centers, and \$2.827343 for paid homes. It includes non-wage costs and provider margins. Unpaid centers have no residual.

2. **Set a local 2019 price.** I compute the local residual the same way, from a local price and the local labor cost. For centers and for paid homes, I take NDCP 2019 county median full-time weekly prices for infants, toddlers, and preschoolers. I weight counties by Census 2019 population ages 0–4 and age groups by the geography's model children in the setting. NDCP gives the geographic pattern. I keep the NSECE national level, because β and ρ are calibrated at NSECE prices:

$$p_g = \frac{\text{NDCP weekly price}_g}{40} \cdot \frac{\text{NSECE national price}}{\text{NDCP national price}},$$

with both national prices child-weighted over the 50 states and DC. The level factor is 1.080 for centers and 1.366 for homes.

3. **Fill missing prices.** Colorado, DC, and Nevada have no 2019 NDCP price, and Mississippi has no 2019 home price. Each takes the nearest NDCP year in 2016–2022, scaled by the national change in each age group's price. Indiana and New Mexico have no NDCP price in 2016–2022. They take the NSECE 2019 price by Census region and community urban density (high, moderate, rural), weighted by the state's 2010 Census children under 5 by tract urban share. A cell is used when its design-based relative standard error is 0.30 or less and it has at least one design degree of freedom. Otherwise the cell takes the national density price times the region price divided by the national price.

4. **Subtract the local labor cost.** The labor cost is local wages times local staff hours per child-hour, summed over the two education groups. For centers, one factor μ_g scales both national tier residuals, so the local tier prices average to the local center price at the P^0 tier mix. The home residual is the remainder:

$$\mu_g = \frac{p_{g,\text{center}} - \text{labor cost}_{g,\text{center}}}{\bar{\delta}_{\text{US,center}}}, \qquad \bar{\delta}_{g,\text{home}} = p_{g,\text{home}} - \text{labor cost}_{g,\text{home}},$$

where the center labor cost and national center residual are averages over the two tiers at the P^0 tier mix. μ_g runs from 0.021 (South Dakota) to 1.992 (DC), median 0.963. The home residual is 0.076 (Mississippi) to 2.825 (DC) times the national one, median 0.918. The build stops if a residual is zero or less.

5. **Apply annual inflation.** CPI increases the nominal residual over time, consistently with the national model: $\bar{\delta}_{g,y} = \bar{\delta}_{g,2019}\,\text{CPI}_y/\text{CPI}_{2019}$.

4.4 Size the care-worker labor market

I use the national wage equation with local inputs. Required staff hours $L_{g,e,y}$ are the model's care hours in each sector times the local staff hours per child-hour, summed over sectors. With G_y national nominal hourly-wage growth since 2019, the worker wage is

$$w_{g,e,y} = w_{g,e,2019}\,G_y\,\frac{A_{g,e,y}}{A_{g,e,2019}}\left(\frac{L_{g,e,y}}{L_{g,e,2019}}\right)^{1/\eta_e},$$

with $\eta_{\text{no BA}} = 4$ and $\eta_{\text{BA}} = 2$, and $L_{g,e,2019}$ at the fitted 2019 baseline. Baseline staff hours command the local wage; increased care demand raises wages according to the national labor-supply elasticities.

5 Parents' employment paths

1. **Estimate relative local employment rates.** For each sex and fine age cell (16–17, 18–19, 20–24, and five-year bins from 25–29 through 55–59), I pool two adjacent ACS years and calculate the person-weighted employment rate. I divide the local rate by the national rate from the same pool. Pooling reduces noise in small local cells.

2. **Apply the national projection.** I multiply the corresponding national CPS/CBO projected employment rate by this local-to-national ratio. Observed ratios run through the 2023–2024 pool; I hold that last ratio fixed afterward. The path combines the national forecast with each place's latest relative position.

3. **Translate the path to caregiver targets.** I combine fine age cells into the model's under-25, 25–34, 35–44, and 45+ groups, separately by sex, using the geography's fixed 2019 ACS age-population weights. The 45+ path uses the available 45–59 projection cells. Each group's target is its 2019 employment-forest rate times the growth in its projected local rate since 2019, capped below one. In each baseline year I fit the employment shift on working options to these targets and hold it fixed for that year's policy comparisons.

6 Implementation and checks

1. **Inputs built.** Demand and supply inputs cover all 52 geographies. The staffing audit has 312 unweighted geography/setting/age records: five center age points and one home input per geography. Of these, 31 change a 2017 value, 7 fill a home value that the 2017 study did not have, and 274 confirm the 2017 value. The center bands (220 unweighted rows, all read in the cited source) must agree with the audited points. Historical-source and hash checks pass. All 52 annual wage paths match their source data.

2. **Baseline fitting.** At local prices before the fit, the model returns P^0; after the fit, care hours match the targets to relative gaps below 10^{-8}. NY minus NYC differs from a separately built rest-of-state target by 1.69 million annual unpaid-center child-hours (1.98% of unpaid-center hours, 0.265% of all rest-of-state care hours), with equal child counts. NYC's own income-rank calibration explains 1.00 million hours; part-time shares and aggregation explain 0.69 million. Paid homes (1.82%) and other paid care (−1.43%) also exceed the strict hours-additivity limit.

3. **Price validation.** I compare model full-time weekly prices with NDCP county median prices, including NDCP's filled-in values. County prices are weighted by Census population ages 0–4, and model prices by local predicted paid-center enrollment across tiers and ages. Model hourly prices become 40-hour weeks. Model ages 0–1, 2, and 3–4 match NDCP's infant (0–23 months), toddler (24–35), and preschool (36–54) groups. Model four-year-olds also include months 55–59, and NDCP's full-time definition varies by state.

   The local residual sets the 2019 model center price from NDCP, so the 2019 comparison is a check of the build, not a test. It covers the 46 states with an NDCP 2019 price in all three age groups: the unweighted correlation is 1.00, and the model price is 8.0% above NDCP in every state, the NSECE level factor. With the national residual on the same states, the correlation is 0.85 and the child-weighted mean model price is 8.8% above NDCP.

   The 2020–2022 comparisons hold the 2019 local residuals fixed and grow them with CPI. They cover 46, 41, and 45 states. The unweighted correlations are 0.987, 0.980, and 0.977, and weekly RMSE is \$23.28, \$24.20, and \$33.30, against \$38.34, \$41.09, and \$46.93 with the 26 October inputs. The mean ratio of model to NDCP price is 1.107, 1.102, and 1.131.

4. **Simulation validation.** The check uses a 1% ACS sample for 2026–2027 across all 52 geographies: 104 geography-year solves. A solve must converge, have a paid-sector price gap below \$0.01 per child-hour, and miss each employment target by less than 0.001 percentage points. The 2 November run, with the local residual, Mississippi's home capacity of 8, and the 26 October calibration, passed all 104 solves. The largest paid-sector price gap was \$0.00036 per child-hour and the largest employment miss was 0.00018 percentage points.

6.1 Open items

- Nevada's home capacity with an infant present is not set by its rule.

- The NY/NYC hours diagnostic remains flagged as described above.

The five items below come from the 2 November review. The wage, staffing, and CCDF figures come from the two supply comparison tests in `state_data/tests`, one local and one on the cluster, with outputs in `state_data/out`. The cluster test first reproduces the national NSECE wages and residuals.

1. **Baseline preferences (3.4).** The ratio fit is in place and matches local care hours and employment. β and ρ stay national, because local data give care totals only, not how families respond to price or income. Open: the 2019-to-2026 drift check has not run on the ratio fit. Resolution: the drift jobs on the 2 November 1% run, queued on the cluster, use the run's own 1% sample; their result replaces the drift figures.

2. **Care-worker wages (4.1).** Open. By Census region, with state wages weighted by children ages 0–4, my no-BA wages are within about 11% of NSECE center-staff wages: higher in the Northeast, lower in the West. The NSECE BA cells are small, and the differences run 14% to 17% in both directions. Resolution: compute NSECE regional wages by education with design standard errors, as for the regional prices, and compare them with my wages at the same staff weights. Where a region differs by more than its standard error, I will decide with the team whether to scale the OEWS ratios to the NSECE regional level, which keeps the OEWS pattern within each region.

3. **Local non-labor residual (4.3).** Resolved. The 2020–2022 price comparison and the 1% simulation check pass with it and with Mississippi's home capacity of 8. South Dakota's center residual is 2.1% of the national one, and Mississippi's home residual is 7.6%, because their NDCP prices are close to their labor costs. Their 2026 equilibrium prices are positive in every sector: \$4.16 low-price center, \$4.62 high-price center, and \$4.28 home in South Dakota; \$2.61, \$5.17, and \$2.94 in Mississippi. South Dakota's two center tiers are only \$0.46 apart, because its small residual leaves mostly the labor-cost difference.

4. **Staffing levels (4.2).** Open; needs a decision. NSECE centers have 4.5 full-time-equivalent children per full-time-equivalent staff member, against 8.9 under the legal ratios; paid homes have 5.05 against 6.1. The ratio of observed to legal children per staff member runs from 0.48 to 0.96 across region and age cells. Observed staffing follows the legal rule less than one for one: across regions within an age group, the log slope is 0.21 (standard error 0.13), and across ages within a region it is 0.68. Resolution: replace the staffing index I_g with I_g^b, with b between 0.2 and 0.7; I propose $b = 0.5$. The national level stays at NSECE. After the decision, I rebuild the supply inputs and rerun the checks. With the local residual, b does not change 2019 prices; it changes the split between labor and non-labor cost, and so how prices respond to wages.

5. **CCDF (4.2).** Resolved: CCDF stays a diagnostic. In CCDF FY2019, 84% of CCDF-subsidized children in home-based care are in licensed homes, against 26% of all paid home-based children in NSECE. The state share has a correlation of 0.05 with selected home capacity. CCDF covers subsidized children only, so it does not measure the licensed share of all paid home care. No input changes.

`SOURCES.csv` lists source links and dates. The staffing evidence register records the rule, effective date, selected value, and conditions.